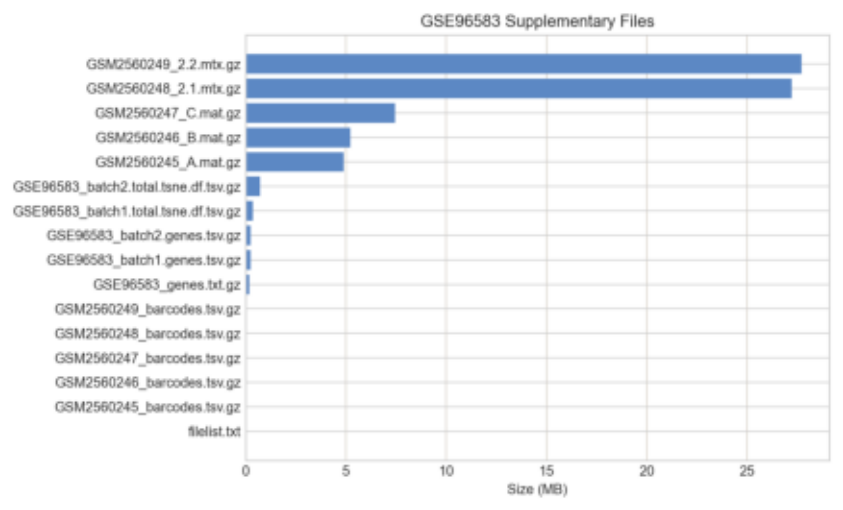
SECTION 1

Data Source: GSE96583

A single PBMC benchmark with labels, batch shift, and condition shift is enough for a defensible milestone 2 story.

- GSE96583 is public on GEO and directly downloadable from the accession page plus the supplementary FTP directory.
- batch1 contains samples A, B, and C, while batch2 contains ctrl and stim.
- This gives us both technical and biological shift without depending on a second dataset.

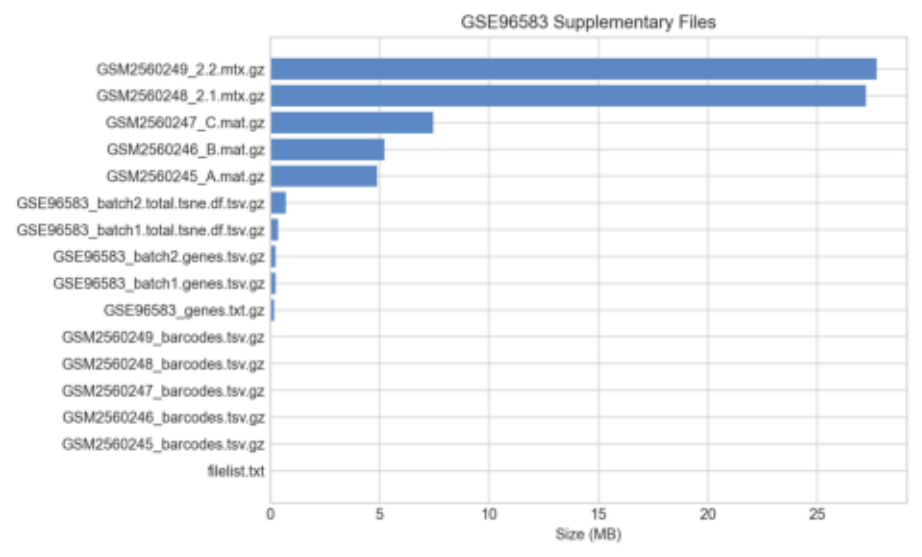


SECTION 2

Raw File Structure

The GEO release is fragmented: matrices and barcodes live in the tarball, but the metadata needed for labels and strict doublet filtering lives outside it.

- batch1 matrices are GSM2560245_A, GSM2560246_B, and GSM2560247_C.
- batch2 matrices are GSM2560248_2.1 (ctrl) and GSM2560249_2.2 (stim).
- Gene tables and tsne.df metadata must be fetched separately before any valid wrangling can start.

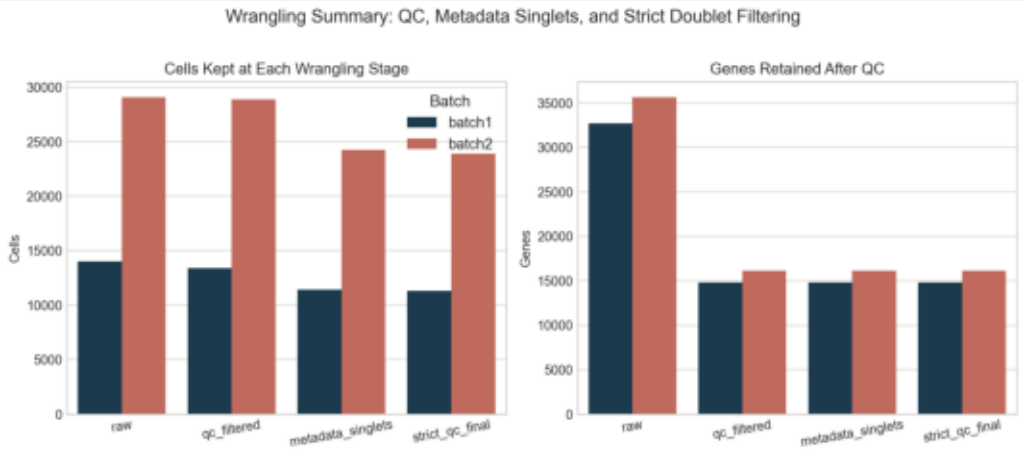


SECTION 2

Data Wrangling Pipeline

After metadata alignment, basic QC, metadata singlets, and strict residual doublet filtering, batch1 keeps 11,308 cells and batch2 keeps 23,906 cells.

- QC keeps cells with at least 200 detected genes and genes seen in at least 3 cells.
- We first keep GEO metadata singlets, then run Scrublet to catch residual doublets.
- We also trim conservative extreme count / gene tails that look doublet-like.
- Cross-batch EDA then aligns both batches on 14222 shared genes.

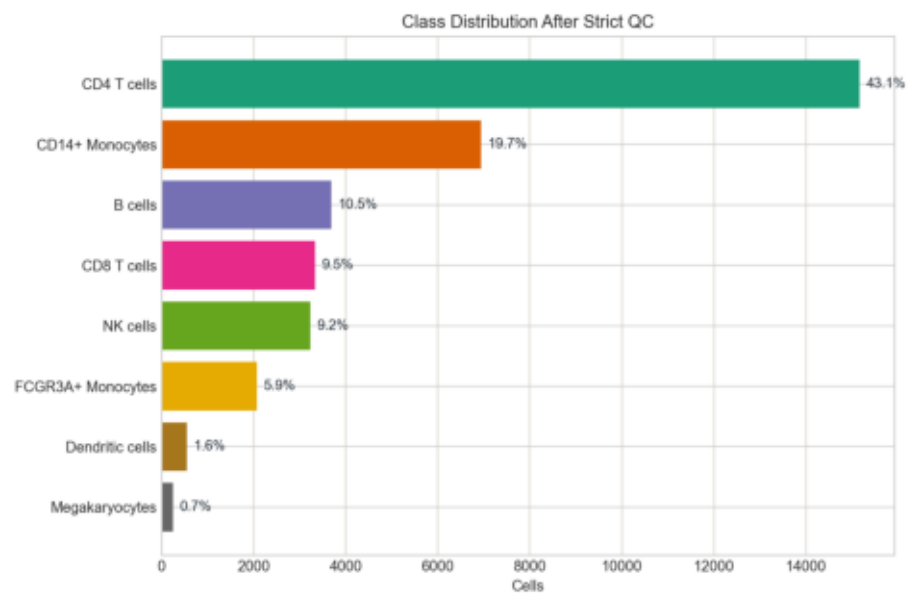


SECTION 2

Class Imbalance

The processed benchmark preserves the main immune classes, but rare classes are still scarce enough to matter for evaluation.

- CD4 T cells and monocytes dominate the benchmark.
- Rare classes under 2% are Dendritic cells, Megakaryocytes.
- This is why later modeling should emphasize macro-F1 and class-wise behavior.

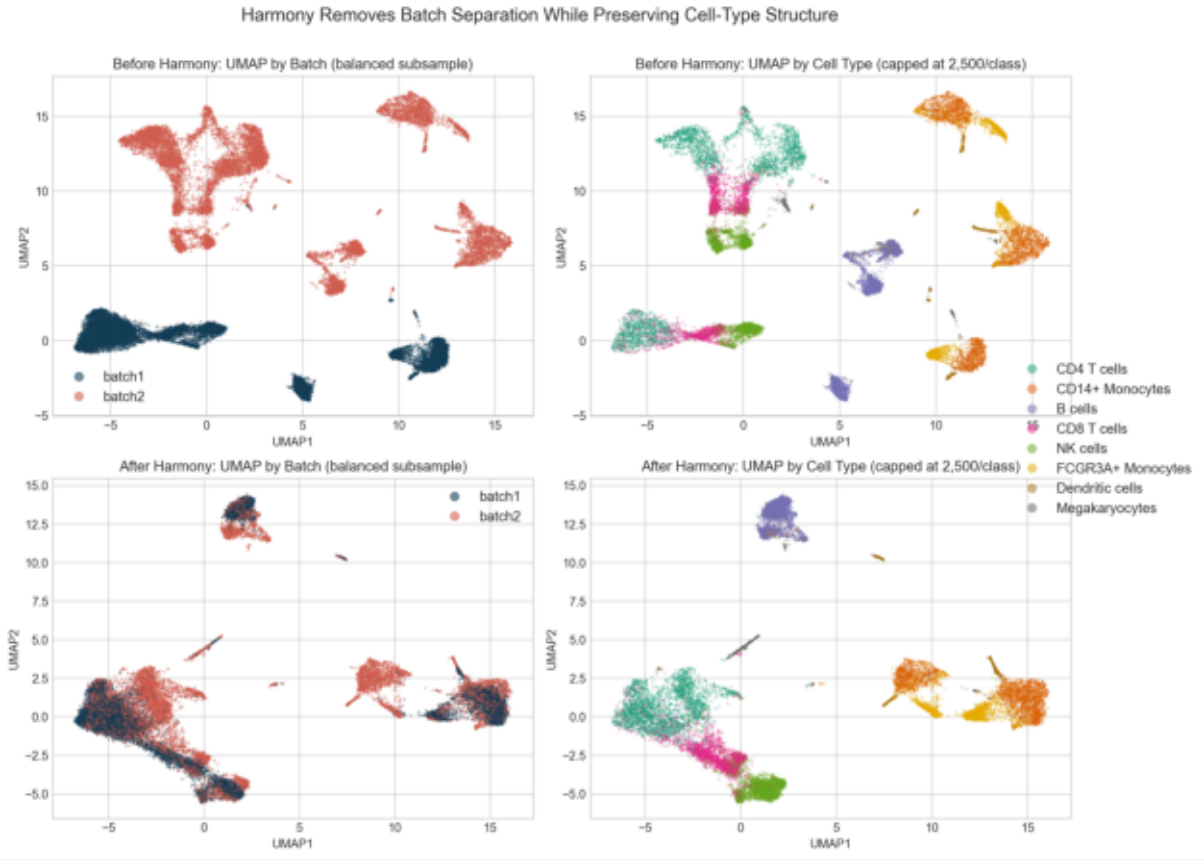


SECTION 2

Batch Effect Before Correction

The uncorrected shared-gene embedding is too batch-driven, so batch correction is justified inside milestone 2.

- Before Harmony, silhouette_batch is 0.0976.
- The pre-correction UMAP shows large batch-driven separation.
- We want the corrected representation to mix batches without destroying cell-type structure.

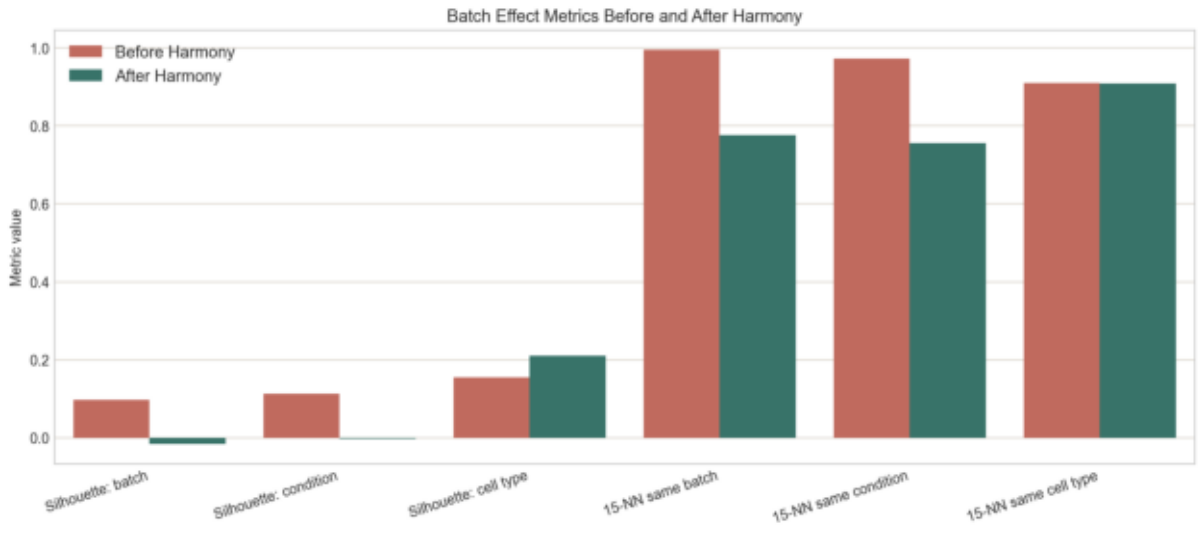


SECTION 2

Harmony Correction

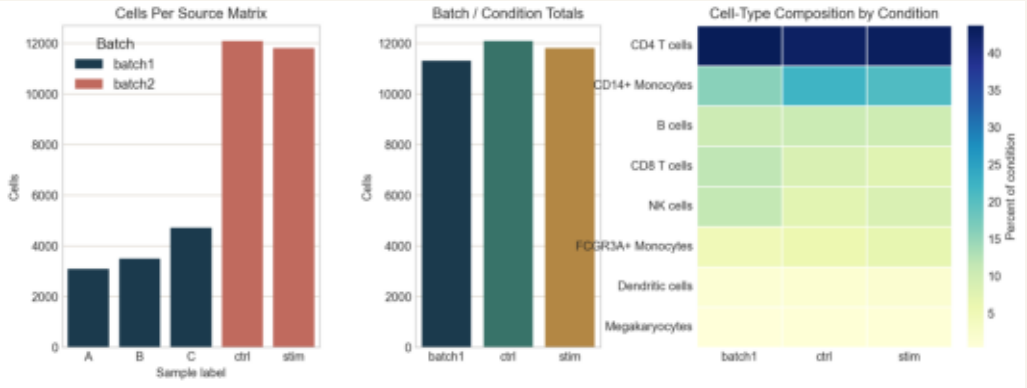
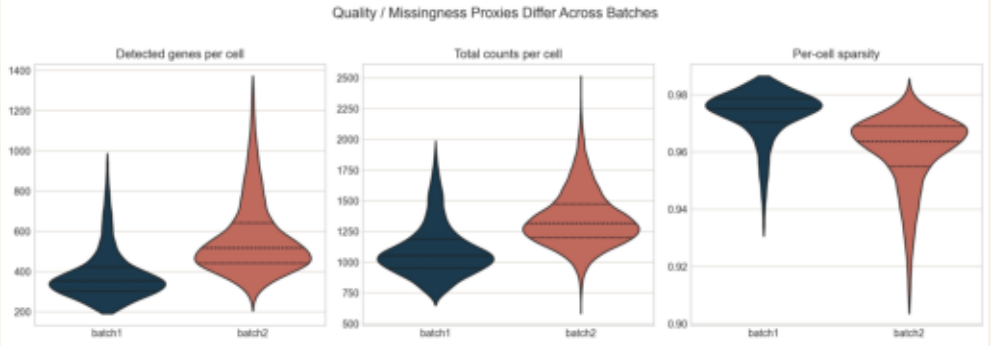
Harmony strongly reduces batch structure while preserving biological neighborhoods.

- Harmony is widely used for batch correction and is especially well known from the R / Seurat workflow.
- silhouette_batch drops from 0.0976 to -0.0163.
- silhouette_cell_type improves from 0.1554 to 0.2106.



SECTION 2

Composition and QC Context

Even after batch correction, the benchmark still contains real condition structure and QC differences.

- batch2 is nearly balanced between ctrl and stim.
- Cell-type composition differs across batch1, ctrl, and stim.
- QC differences are part of the realism of the benchmark and should be described rather than hidden.